

Unit - 6(Reproduction Unit) - 200 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



Good evening
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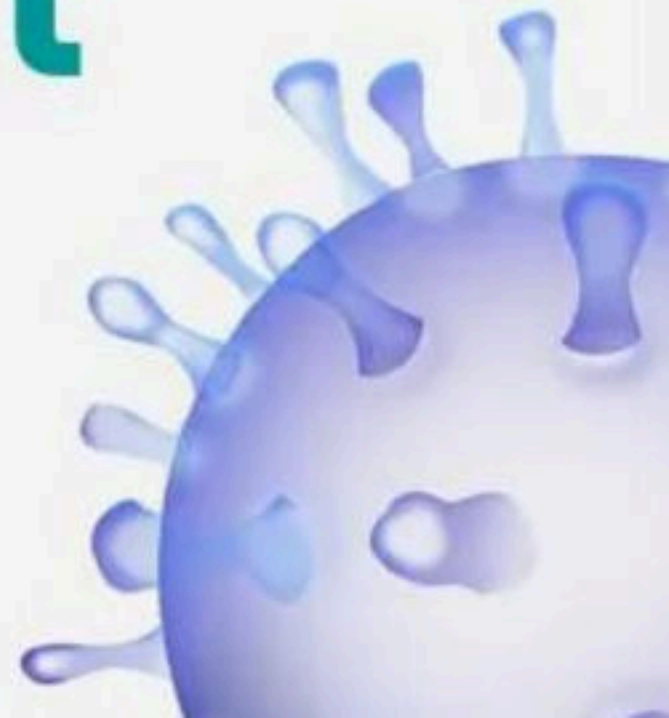


200 MCQs

Reproduction Unit

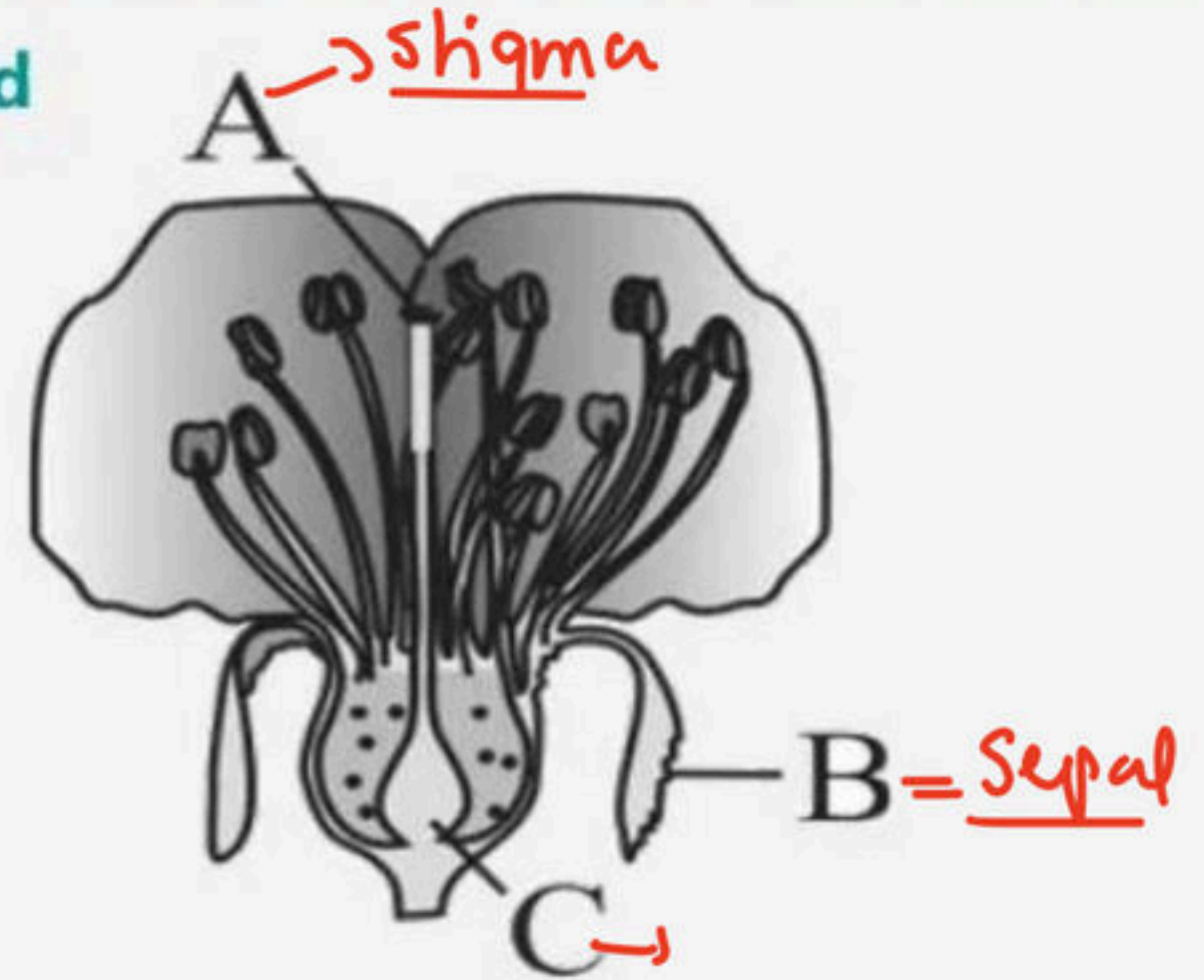


BIOLOGYLIVE



1. The following figure represents a ^{Bisexual} flower and identify A, B, C.

- (1) Unisexual A - stigma, B - sepal, C - ovary
- ✓ (2) Bisexual A - stigma, B - sepal, C - ovary
- (3) Bisexual A - stigma, B - petal, C - ovary
- (4) Unisexual A - stigma, B - petal, C - ovary



2. The technique advised by a doctor to overcome the problem of infertility :- ART

(1) RCH

(2) MTP

(3) RTI

✓ (4) ART



3. Which one of the following is an example of "hormone" releasing IUD?

(1) CuT

✓ (2) LNG 20

(3) Cu7

(4) Multiload 375



4. Which of the following disease are not completely curable?

- (1) Hepatitis B ✓
- (2) HIV ✓
- (3) Genital herpes ✓
- ✓ (4) All the above



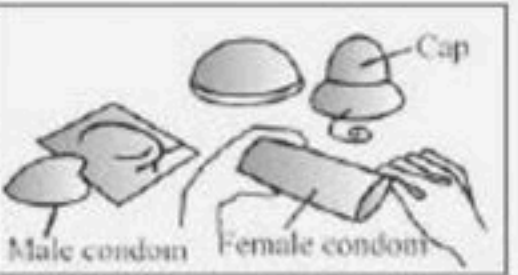
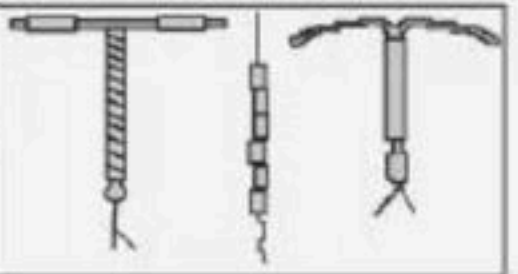
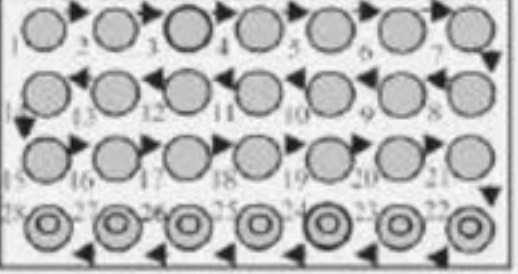
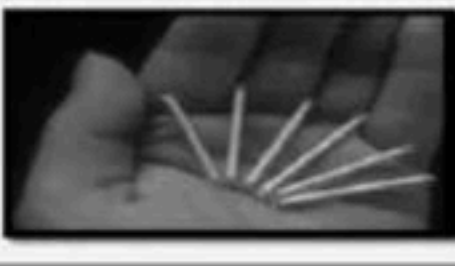
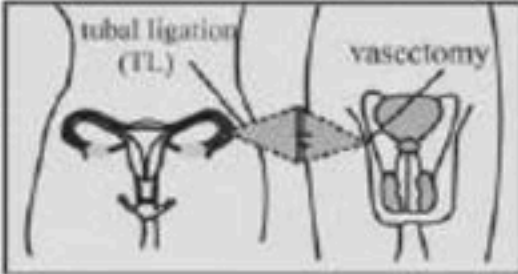
5. Match the following columns based on the figures given

(1) A - P; B - R; C - Q; D - S; E - T

✓ (2) A - P; B - Q; C - R; D - S; E - T

(3) A - P; B - Q; C - S; D - R; E - T

(4) A - T; B - Q; C - R; D - S; E - P

Column I	Column II
A. <u>Barriers</u> →	P. 
B. <u>Intra uterine devices</u> →	Q. 
C. <u>Oral contraceptives</u> →	R. 
D. <u>Injectables</u> →	S. 
E. <u>Surgical methods (Sterilization)</u> →	T. 



6. Which of the following is a non- invasive technique for monitoring foetal growth and detecting congenital diseases?



- ✓ (1) Ultrasonography
- (2) Amniocentesis
- (3) Chorionic villus sampling
- (4) All of these



7. Which of the following does not give the defined action of contraceptive?

- (1) Hormonal contraceptives-prevent/ retard entry of sperms, prevent ovulation and fertilization.
- ✓ (2) Vasectomy- ~~prevents~~ spermatogenesis $\rightarrow$ transport X
- (3) Barrier methods- prevents fertilization
- (4) Intra uterine devices-increase phagocytosis of sperms, suppress sperm motility and fertilizing capacity of sperms.



8. Copper-T acts by :-

- (1) Suppress sperm motility and the fertilising capacity of sperms
- (2) Suppression of fertilization by forming a membrane
- (3) Acting as a barrier
- (4) Makes cervix hostile to the sperms.



9. The diaphragm, cervical cap and vaults are

(1) Disposable chemical barriers

✓ (2) Reusable contraceptives

(3) Non-medicated IUDs

(4) Cu-releasing IUDs



10. Which of the following is correct?

Female.

- (1) In India, often the ~~male~~ female is blamed for the couple being childless
- (2) Sometimes, use condoms during coitus. **X**
- (3) Government of India legalised MTP in 1961 with some strict conditions to avoid its misuse. **X**
- ✓ (4) ICSI is a procedure to form an embryo in the laboratory in which a sperm is directly injected into the ovum.



11. Lippes loop is a/an

- (1) Hormone releasing IUD
- (2) Copper releasing IUD
- ✓ (3) Non- medicated IUD
- (4) Skin implant for sustained release of gonadotropins



12. Coitus interruptus/withdrawal method concerns with

- ✓ (1) Withdrawal of penis from vagina before ejaculation
- (2) Withdrawal of penis from vagina after ejaculation ✗
- (3) Prevention of coitus
- (4) Coitus after fertile period



13. Lippe's loop is a type of contraceptive used as:

(1) Vault barrier



✓ (2) Non-Medicated IUD

(3) Copper releasing IUD

(4) Cervical barrier



14. The sexually transmitted disease, that can affect both the male and the female genitals and may damage the eyes of babies born of infected mothers is:

(1) AIDS

(2) Syphilis

✓ (3) Gonorrhoea

(4) Hepatitis



15. These statements are given regarding MTP. Choose which one is the correct option. :-

- ✓ I. Are generally advised during first trimester
- II. Are used as contraceptive method X
- III. Are always surgical X
- ✓ IV. Require an assistance of qualified medical personnel

(1) I and III

(2) II and III

✓ (3) I and IV

(4) III and IV



16. India's population crossed 1 billion in :-

(1) Dec. 1999

(2) May 2001

(3) Dec. 1991

✓ (4) May 2000

↓
may 2000



17. What is the main factor affecting the population growth?

(1) Increase in health care facilities ✓

(2) Better quality of life. ✓

✓ (3) Both 1 & 2

(4) Industrialization



18. Copper releasing IUD is the same kind of contraceptive as

(1) Diaphragm

✓ (2) Multiload 375

(3) LNG - 20

(4) Mirena



19. A weekly contraceptive pill is



(1) Mala A

✓(2) Saheli

(3) Mala D

(4) Saheli-2



20 Choose the right one among the statements given below:-

- (1) IUDs are generally inserted by the user herself
- ✓ (2) IUDs increase phagocytosis reaction in the uterus
- (3) IUDs suppress gametogenesis
- (4) IUDs once inserted need not be replaced



21. Match the column-I with column-II.

Column I		Column II	
A	Transfer of ovum collected from a donar	P	Gametes fallopian transfer
B	Semen introduced into vagina directly	Q	Intrauterine transfer
C	Embryo with 8 blastomeres	R	Artificial insemination
D	Sperm is directly injected into ovum	S	Intra cytoplasmic sperm iniecton

(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

☒ (3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



22 Fill in the blanks

Oral contraceptive pills have to be taken daily for a period of ²¹___A___ starting preferably within the first ___B___ of menstrual cycle. After a gap of ___C___ days (during which menstruation occurs) it has to be repeated in the same pattern till the female desires to prevent contraception.

- ✓ (1) A - 21 days; B - 5 days; C - 7 days
- (2) A - 28 days; B - 7 days; C - 5 days
- (3) A - 28 days; B - 5 days; C - 7 days
- (4) A - 22 days; B - 7 days; C - 5 days



23. Primary steps that can be taken towards attaining reproductive health are

- (1) Counselling and creating awareness among people about reproductive organs, adolescence and associated changes ✓
- (2) Safe and hygienic sexual practices ✓
- (3) Creating awareness about STDs including AIDS, better detection and cure of STD ✓
- (4) All of them ✓



24. Which of the following contraceptive methods do involve a role of hormone?

- (1) Lactational amenorrhea, Pills Emergency contraceptives. ✓✓
- (2) Barrier method, Lactational amenorrhea, Pills. ✗
- (3) CuT, Pills, Emergency contraceptives. ✗
- (4) Pills, Emergency contraceptives, Barrier methods. ✗



25. Diaphragms are contraceptive devices used by the females. Choose the correct option from the statements given below:

- i. They are introduced into the ~~uterus~~ ^{cervix} X
- ii. They are placed to cover the cervical region ✓
- iii. They act as physical barriers for sperm entry ✓
- iv. They act as spermicidal agents X

(1) and ii

(2) i and iii

✓ (3) ii and iii

(4) iii and iv



26. Which one is a method of birth control?

(1) IVF-ET

(2) HTF

(3) GIFT

✓ (4) IUDs



27. The complication of the STDs includes.

✓ a. PID.

✓ b. Abortionsc.

✓ c. Still birthd.

✓ d. Ectopic pregnanciese.

✓ e. Infertility

✓ f. Cancer of reproductive tract

(1) a, c and d

(2) a, b, d and e

(3) b, c, d, e and f

✓ (4) All of the above



28. Progestasert and LNG-20 are :-

- (1) Implants
- (2) Copper releasing IUDs
- (3) Non-medicated IUDs
- ☒ (4) Hormone releasing IUDs



29 Administration of which of the following within 72 hours of coitus have been found to be effective as emergency contraceptives?

i. Progestogen ✓

ii. Progestogen-estrogen combinations ✓

iii. Intrauterine devices ✓

iv. Diaphragms or cervical caps

(1) i and ii only

(2) i, iii and iv

(3) ii and iii only

✓ (4) i, ii, and iii



30. Intensely lactating mothers do not generally conceive due to

- (1) Hyper secretion of gonadotropins
- ✓ (2) Suppression of gonadotropins
- (3) Suppression of fertilization
- (4) Suppression of gameteic transport



31. CDRI stands for

- (1) Central Drug Refund Institute
- (2) Central Drug Related Institute
- ✓ (3) Central Drug Research Institute ✓
- (4) Children Development Research Institute



32. Identify the correct statements.

I. There are many side effects of tubectomy and vasectomy. ✗

II. Purpose of tubectomy is to prevent egg and sperm fusion. ✓

III. The most important component of the oral contraceptive pills is progesterone. ✓

IV. Contraceptive oral pills help in birth control by preventing ovulation. ✓

(1) I, II and III

(2) I and II

✓ (3) III, IV and II

(4) II and IV



33 Identify the correct statements about lactational amenorrhea.

I. Ovulation does not occur during the lactational period. ✓

II. Side effects are huge. ✗

III. Chances of the failure of contraception are almost nil up to six years following parturition. ✗

IV. It is a natural method of contraception. ✓✓

V. It increases phagocytosis of sperms. ✗

(1) I, III, II and V

(2) All except II

✓ (3) I, IV

(4) All of these



34. In the rhythm method of birth control, the couple must abstain from coitus at least

- (1) From one day before ovulation till one day after ovulation
- (2) From two days before ovulation till two days after ovulation
- ✓ (3) From three days before ovulation till three days after ovulation
- (4) From ten days before ovulation till ten days after ovulation

$$\begin{array}{c} \underline{10} - \underline{17} \\ \downarrow \\ -3 \leftarrow 14 \rightarrow +3 \end{array}$$



45 - 50

35. Nearly ___ A ___ million MTPs performed in a year all over the world which accounts to ___ B ___ of the total number of conceived pregnancies in a year.

 $\frac{1}{5}^{\text{th}}$

(1) A - 40 to 50 , B - $\frac{1}{4}^{\text{th}}$

(2) A - 45 to 50 , B - $\frac{1}{4}^{\text{th}}$

✓ (3) A - 45 to 50 , B - $\frac{1}{5}^{\text{th}}$

(4) A - 40 to 50 , B - $\frac{1}{5}^{\text{th}}$



36. Which one of the following cannot be considered as Assisted Reproductive Technology?

- (1) Zygote Intra fallopian tube transfer (ZIFT)
- (2) Gamete Intra Fallopian Transfer (GIFT)
- ✓ (3) Medical Termination of Pregnancy (MTP)
- (4) Intra Cytoplasmic Sperm Injection (ICSI)



37. Amniocentesis is a technique used to detect

- (1) Down syndrome
- (2) Haemophilia
- (3) Sickle cell anaemia
- (4) All the above



38. Saheli, developed by Indian scientists is a

- (1) Steroidal preparation
- (2) Hormonal preparation
- ✓ (3) Non-steroidal preparation
- (4) Effective preparation to demobilise the sperms



39 Assertion : Intra uterine devices (IUD) are very effective Contraceptive method.

Reason : IUD don't allow sperms to enter the uterus.

- ☒ (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



40. Emergency contraceptives could be used to avoid possible pregnancy due to

- (1) MTP
- (2) Rape ✓
- (3) Casual unprotected intercourse ✓
- (4) ✓ Both 2 and 3



41. Match the following and choose the correct option

Column I		Column II	
A	Hepatitis	P	Infection of reproductive system
B	Ectopic pregnancy	Q	7' April, 1984
C	PID	R	Fallopian tubes <u>implantation</u>
D	World Health day	S	Detection of antibody/organisation antigen
E	ELISA technique	T	Hepatitis B virus

- (1) A-T, B-Q, C-P, D-R, E-S ✓ (2) A-T, B-R, C-P, D-Q, E-S
(3) A-T, B-Q, C-S, D-R, E-P (4) A-T, B-R, C-S, D-Q, E-P



42. What is the reason for popularity of barrier method of family planning?

- (1) The absence of side effect ✓
- (2) Comparatively more reliable
- (3) Protection from sexually transmitted diseases ✓
- ✓ (4) Both (1) and (3)



43. Following statements are given regarding MTP. Choose the correct options given below:

I. MTPs are generally advised during first trimester ✓

II. MTPs are used as a contraceptive method ✗

III. MTPs are always surgical ✗

IV. MTPs require the assistance of qualified medical personnel ✓

(1) II and III

(2) II and IV

(3) I and IV ✓

(4) I and II



44. MTPs are considered relatively safe during

(1) 24 weeks of pregnancy

(2) 180 days of pregnancy

✓ (3) First trimester / 12 week

(4) Second trimester



45. Which of the following is true? Consider the statements given below regarding contraception :-

- ↓
- ✓ (1) Medical termination of pregnancy (MTP) during first trimester is generally safe
 - (2) Generally chances of contraception are nil until mother breast feeds the infants upto two years.
 - (3) Intra uterine devices like copper-T are non effective contraceptives
 - (4) Contraceptive pills may be taken upto one week after coitus to prevents contraception



46. Which one of these STDs does not specifically affect the sex organs?

- ✓ (1) AIDS
- (2) Gonorrhoea
- (3) Syphilis
- (4) Genital warts



47. Read the following statements and find out the incorrect statement :-

I. Barrier methods are available for both males and females. ✓

II. In barrier methods, ovum and sperms are prevented from physically meeting with the help of barriers. ✓

III. Condoms are made of thin rubber/ latex sheath that are used to cover the penis in male or vagina and cervix in the female, just before coitus so that the ejaculated semen would not enter into the female reproductive tract. ✓

IV. Diaphragm, cervical caps and vaults are barriers made of rubber that are inserted into the female reproductive tract to cover cervix during coitus.

These barriers are disposable. ✗

(1) I and II

(2) II and III

(3) IV ✓

(4) III and IV



48. Which of the following STDs cannot be treated with antibiotics?

- (1) Gonorrhoea
- (2) Chancroid
- (3) Chlamydia
- ✓ (4) Genital herpes →



49 The hormone releasing intrauterine devices such as progestasert specifically :-

- (1) Suppress sperm motility
- (2) Increase phagocytosis of sperms
- ✓ (3) Make uterus unsustainable for implantation
- (4) Suppress fertilizing capacity of sperms



50. Match the ARTs with their descriptioni.

- (i) Collected gametes are made to form the zygote in the laboratory. $\Rightarrow$ IVF
- ii. Zygote or early embryo with upto 8 blastomeres is transferred into the oviduct. $\rightarrow$ ZIFT
- iii. Zygote with more than 8 blastomeres is transferred into the uterus. $\rightarrow$ IUT
- iv. Fusion of gametes within the female. $\rightarrow$ H
- v. Transfer of ovum from a donor to the oviduct of the recipient. $\rightarrow$ GIFT
- vi. Sperm is injected into the ovum in-vitro $\rightarrow$ ICSI

A. GIFT

B. ZIFT

C. AI

D. ICSI

E. IUT

F. IVF

G. IUI

H. In - vivo fertilisation

(1) i-F, ii- B, iii-A, iv- H, v-A, vi - G

(2) i-G, ii - B, iii - F, iv- H, v-A, vi - D

(3) i-G, ii - B, iii- F, iv- H, v - C, vi - E

(4) i - F, ii - B, iii- E, iv- H, v-A, vi - D



51. Typical angiosperm anther is

(1) Unilobed and dithecal

✓(2) Bilobed and dithecal ✓

(3) May be both (1) and (2)

(4) Bilobed and tetralocular



52 Primary endosperm nucleus is formed by fusion of

- (1) Ovum and male gamete
- (2) One polar nucleus and male gamete
- (3) Two polar nuclei and two male gametes
- ✓ (4) Two polar nuclei and one male gamete



53. The coconut water and edible part coconut are
equivalent to

- (1) Endocarp
- ✓ (2) Endosperm
- (3) Mesocarp
- (4) Embryo



54. Testa of the seed is produced from :-

- (1) Hilum
- (2) Ovary wall
- (3) Funicle
- (4) ☒ Outer integument of ovule



55. In a flowering plant, archesporium gives rise to

- (1) Wall and the tapetum
- (2) Only the wall of the sporangium
- ✓ (3) Both ^{Anther} wall and the sporogenous cells
- (4) Only tapetum and sporogenous cells



56. Select the most appropriate statement w.r.t. to wind pollination

(1) Pollen grains are light, non-sticky ✓

(2) Well-exposed stamens ✓

(3) Large often-feathery stigma to easily trap air-borne pollen grains ✓

✓ (4) All of them



57. Starting from the innermost part, the correct sequence of parts in an ovule are,

- (1) egg, nucellus, embryo sac, integument
- ✓ (2) egg, embryo sac, nucellus, integument
- (3) embryo sac, nucellus, integument, egg
- (4) egg, integument, embryo sac, nucellus.



58. What is functional megaspore referred to as?

- (1) The megaspore that degenerates after formation
- ✓ (2) The megaspore that does not degenerate and undergoes mitosis later
- (3) The megaspore that undergoes reduction division.
- (4) The megaspore that is functionally inactive



59. Endosperm is completely consumed during seed formation in

(1) Pea, castor and coconut

(2) Castor, bean and coconut

✓ (3) Pea, groundnut and beans $\Rightarrow$ direct

(4) Groundnut, bean and castor



60. In some plants, stigma and anther mature at different times because:

- (1) It attracts pollinators
- (2) It facilitates self-pollination
- (3) It prevents cross pollination X
- (4) It facilitates cross pollination ✓



61. Filiform apparatus is found in

(1) Secondary nucleus

✓ (2) Synergids

(3) Egg cell

(4) Antipodal cell



62 Ovules are attached to a parenchymatous cushion ↓
called

- (1) Nucellus
- (2) Obturator
- (3) Conducting tissue
- ✓ (4) Placenta



63. In angiosperms the following events occur during microgametogenesis

- ③ i. Mitotic division in microspore ✓
- ① ii. Meiotic division in microspore mother cell ✓
- ④ iii. Formation of pollen grain
- ② iv. Formation of microspores
- v. Formation of mature male gametophyte

The correct sequence of events is

- (1) ii, i, iv, v, iii
- (2) iii, ii, i, v, iv
- ✓ (3) ii, iv, i, iii, v
- (4) iv, v, ii, i, iii



64. Pollen grains

- A. Represent male gametophytic phase of plant**
- B. Can cause severe allergies like asthma and bronchitis ✓**
- C. Are rich in nutrient and are used as food supplements ✓**
- D. Contains two male gametes and vegetative cell ✓**
- E. Are available in form of tablets and syrups in market of western countries ✓**

(1) Only A is correct

✓ (2) All are correct

(3) All are wrong

(4) Only A, B and C are correct



65. In the embryos of a typical dicot and a grass, true homologous structures are:

(1) Coleorhiza and coleoptile

(2) Coleoptile^X and scutellum

✓(3) Cotyledons and scutellum✓✓

(4) Hypocotyl and radicle.



²⁰²⁵
66. Male gametes in angiosperms are formed by :-

- (1) Mitotic division in vegetative cell
- ✓ (2) Mitotic division in generative cell
- (3) Meiotic division in generative cell
- (4) Meiotic division in pollen mother cell



67. From among the situations given below, choose the one that prevents both autogamy and geitonogamy.

- (1) Monoecious plant bearing unisexual flowers
- ✓ (2) Dioecious plant bearing only male or female flowers =>
- (3) Monoecious plant with bisexual flowers
- (4) Dioecious plant with bisexual flowers



68. Statement A : Pollen grains shed at two-celled condition, in such plants, the vegetative cell divides and forms the two male gametes during the growth of pollen tube in the stigma.

Statement B : Emasculated flowers have to be covered with a bag of suitable size, generally made up of butter paper ✓

(1) Statement A is correct but Statement B is incorrect.

✓ (2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.



69. Which of the following is not involved in post fertilization events?

- (1) Endosperm and embryo development
- (2) Maturation of ovule into seed
- (3) Maturation of ovary into fruit
- (4) Formation of gametes → pre-fertiliz.



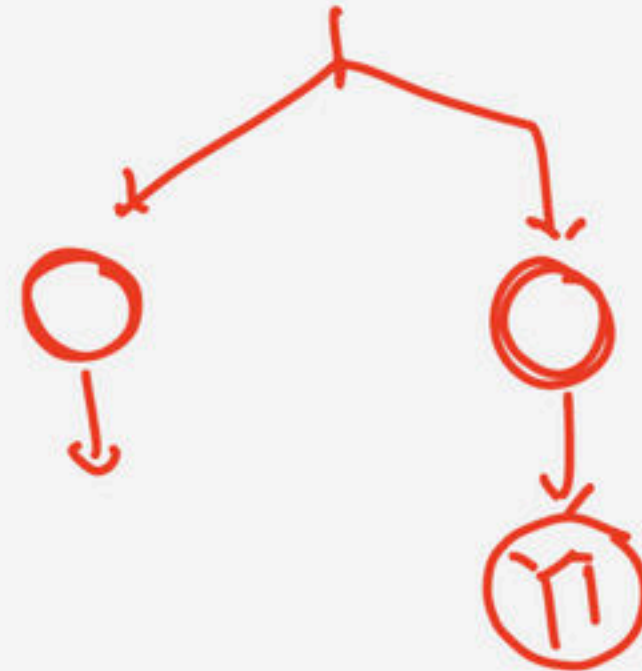
70. Milky water of green coconut is :-

- (1) Liquid chalaza
- (2) Liquid nucellus
- ☒ (3) Liquid endosperm
- (4) Liquid female gametophyte



71. Double fertilization means :-

- (1) Fusion of two polar bodies
- ☒ (2) A male gamete fused with egg and second male gamete fused with secondary nucleus
- (3) Fusion of male gametes and ovum
- (4) All the above



72. Match the columns I and II.

Column I		Column II	
A	Pollen bank →	P	(-196°C)
B	Generative cell	Q	Abundant food reserve
C	Vegetative cell	R	Spindle shaped cell
D	Parthenium →	S	Carrot grass

(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

✓ (3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



73. The plant parts which consist of two generations-one within the other:

- a) Pollen grains inside the anther ✓
- b) Germinated pollen grain with two male gametes ✗
- c) Seed inside the fruit ✗
- d) Embryo sac inside the ovule ✓

(1) (a), (b) & (c)

(2) (c) & (d)

✓ (3) (a) & (d)

(4) (a) only



74. Which one of the following statement is correct?

- (1) Endothecium produces the microspores
- ✓ (2) Tapetum nourishes the developing pollen
- (3) Hard outer layer of pollen is called intine
- (4) Sporogenous tissue is haploid



75. Statement A : The innermost wall layer is the tapetum. It nourishes the developing pollen grains.

Statement B : When the anther is young, a group of compactly arranged heterogenous ^{homolog.} cells called the sporogenous tissue occupies the outer layer of each microsporangium

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



76 Which of the following is the tallest flower?

- (1) Vallisneria
- (2) Lotus
- ✓ (3) Amorphophallus
- (4) Zostera



77. In dicot plants, arrangement of nuclei in normal embryosac is

(1) 3+2+2

☒ (2) 3+2+3

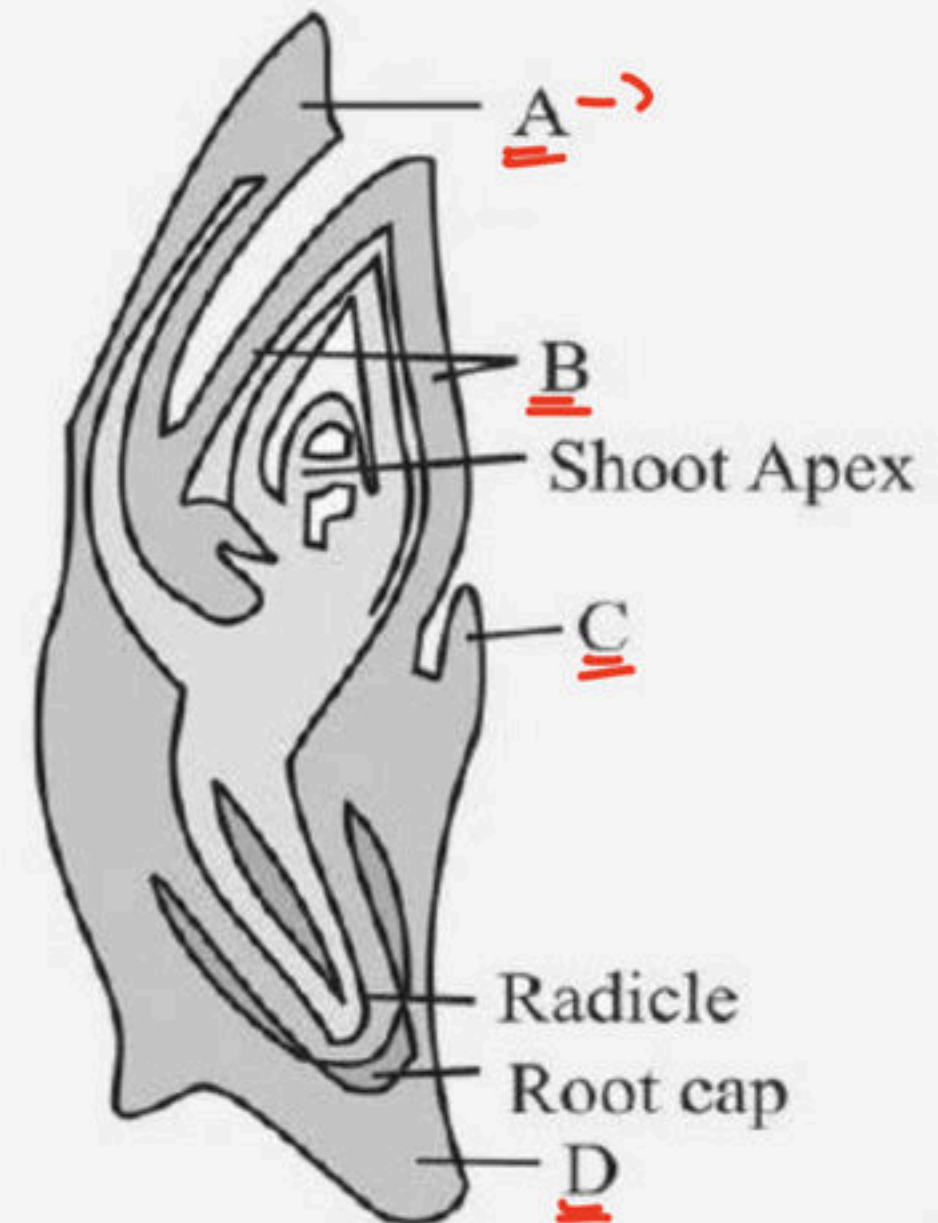
(3) 3+3+3

(4) 2+3+3



78. Choose the correct option for A, B, C and D of a monocot embryo.

- (1) A- Scutellum; B- Epiblast; C- Coleoptile; D- Coleorhiza
- ✓ (2) A-Scutellum; B- Coleoptile; C- Epiblast; D- Coleorhiza
- (3) A- Coleoptile; B- Scutellum; C-Epiblast; D- Coleorhiza
- (4) A- Scutellum; B- Coleoptile; C- Coleorhiza; D- Epiblast



79. Pollination by water takes place in :-

- (1) Vallisneria
- (2) Hydrilla
- (3) Zostera
- (4) All the above



80. In a fertilised embryo sac, the haploid, diploid and triploid structures are:

- ✓ (1) Synergid, zygote and primary endosperm nucleus
- (2) Synergid, antipodal and polar nuclei
- (3) Antipodal, synergid and primary endosperm nucleus
- (4) Synergid, polar nuclei and zygote.



81. Endosperm development ____ embryo development.

✓ (1) Precedes ↗

(2) Follows after

(3) Is not required

(4) Occurs after seed is formed



82.* Which of the following statements is wrong?

(1) Pollen grains remain viable for several months ✗

because their outer covering is made of sporopollenin

(2) No enzyme can degrade sporopollenin ✓

(3) Pollen grains are well represented in fossil strata due to sporopollenin ✓

(4) Pollen wall has cavities containing proteins ✓



83. The record of oldest seed dormancy is of

- (1) 2,000 years
- (2) 15,000 years
- ✓✓ (3) 10,000 years
- (4) 12,000 years



84. In flower the whorl of stamens represents

(1) Corolla

(2) Calyx

✓ (3) Androecium

(4) Gynoecium



85. Development of seedless fruit in an unfertilized flower is

- (1) Polyembryony
- (2) Sporophytic budding
- ✓ (3) Parthenocarpy
- (4) Micropropagation



86. Through which route does the pollen tube enter the ovule :-

- (1) Chalaza ✓
- (2) Integument ✓
- (3) Micropyle ✓
- ✓ (4) All of these



87. In anther culture, the androgenic haploid plants are
obtained from :- Young pollen grain

- (1) Anther wall
- (2) Anther tapetum
- (3) Connective tissue
- ✓ (4) Young pollen grain



88. Which of the following statement is true?

i. Pollen release and stigma receptivity are always synchronized in all species. ✗

ii. Pollination does not guarantee the transfer of right type of pollen. ✓

iii. Micropyle represents the basal part of ovule. ✗

iv. Most zygotes divide only after certain amount of endosperm is formed. ✓

(1) i. iii

(2) i. ii, iv

✓ (3) ii. iv

(4) ii. iii. iv



89. Male gametophyte of angiosperm is :- Microspore

- ✓ (1) Microspore
- (2) Microsporangium
- (3) Nucellus
- (4) Stamen



90. Embryo develops at ["]which end ["] of the embryosac :-

(1) Chalazal end → Antipodal cells

✓✓ (2) Micropylar end

(3) Both (1) and (2)

(4) None of the above



91. In the anther wall

- (1) Tapetum lies next to the epidermis ✗
- ✓ (2) Middle layers lie between endothecium and tapetum
- (3) Tapetum lies just inner to the endothecium
- (4) Endothecium lies inner to the middle layers



92. In angiosperms the following events occur during microgametogenesis

- i. Mitotic division in microspore
- ii. Meiotic division in microspore mother cell ✓
- iii. Formation of pollen grain ✓
- iv. Formation of microspores ✓
- v. Formation of mature male gametophyte ✓

The correct sequence of events is

- | | |
|-------------------------|-----------------------|
| (1) ii, i, iv, v, iii | (2) iii, ii, i, v, iv |
| ✓ (3) ii, iv, i, iii, v | (4) iv, v, ii, i, iii |



93 Statement A : Tiny seeds are produced by Ficus, Orobanche and Striga.

Statement B : In many Citrus and Mango varieties some of the nucellar cells surrounding the embryo sac start dividing, protrude into the embryo sac and develop into the embryos.

(1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

✓ (3) Both statements are correct.

(4) Both statements are incorrect.



94. Which characters should be possessed by autogamous flowers?

- (1) Synchrony in pollen release and stigma receptivity ✓
- (2) The anthers and the stigma should lie close to each other ✓
- ✓ (3) Both 1 and 2
- (4) The anthers and the stigma should lie on different flowers



95. In a leaf cell of angiosperm contain 50 chromosome,
the number of chromosome in egg cell, pollen grain,
endosperm and integument would be

(1) 25, 25, 50, 50

✓ (2) 25, 25, 75, 50

(3) 25, 25, 25, 50

(4) 25, 50, 50, 75



96. Pollen grains are rich in nutrients. It has become a fashion in recent years to

- (1) Consumed in the form of tablets and syrups ✓
- (2) Pollen consumption has been claimed to increase the performance of athletes and race horses ✓
- (3) use pollen tablets as food supplements. ✓
- (4) ✓ All of these





97 Autogamy can occur in a chasmogamous flower if:

- (1) Pollen matures before maturity of ovule
- (2) Ovules mature before maturity of pollen
- ✓ (3) Both pollen and ovules mature simultaneously
- (4) Both anther and stigma are of equal lengths.



98. Amongst the pollinating agent which one is the most dominant?

(1) Butterflies and flies

✓✓ (2) Bees

(3) Beetles and wasps

(4) oths and birds



99. Assertion : Some fruits are seedless or contain non-viable seeds.

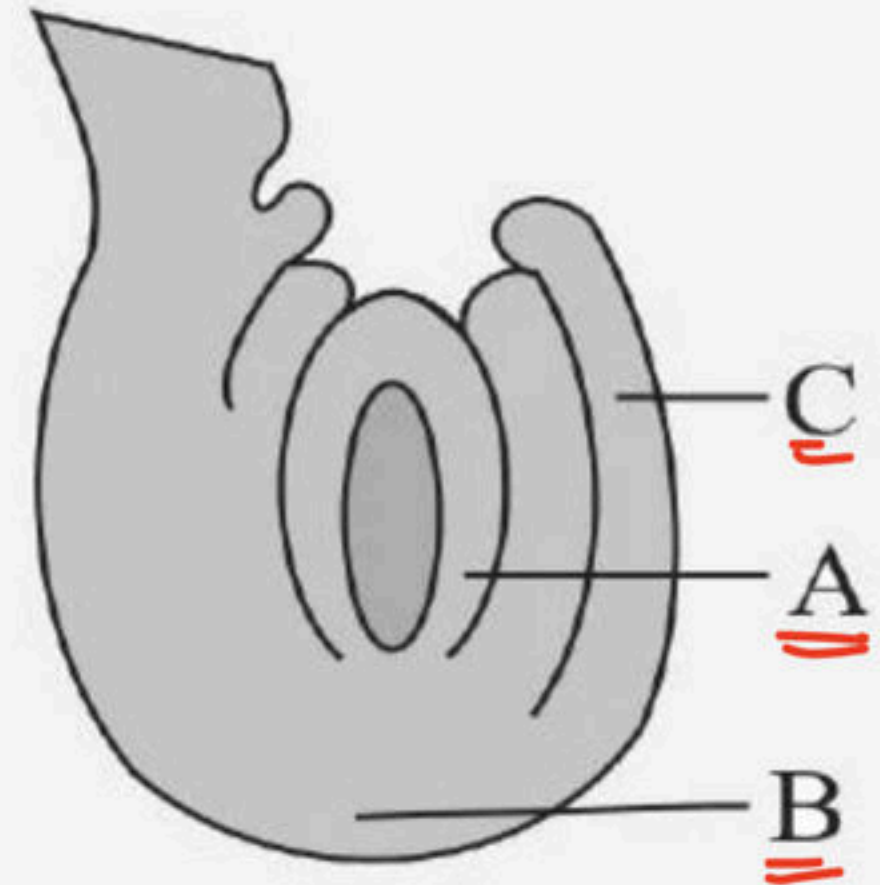
Reason : They are produced without fertilisation.

- ✓ (1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false.



100. Identify A, B, C

- (1) A - Nucellus, B - Micropyle, C - Outer integument
- ☒ (2) A - Nucellus, B - Chalazal end, C - Outer integument
- (3) A - Nucellus, B - Chalazal end, C - Inner integument
- (4) A - Nucellus, B - Micropyle, C - Inner integument



101. Polar body is formed during the formation of

- (1) Primary oocyte
- ✓ (2) Secondary oocyte
- (3) Oogonium
- (4) Tertiary follicle



102. Match the months listed in Column - I write the organogenesis of foetus in Column - II

Column I		Column II	
A	First month	P	Separation of eye lids
B	Second month	Q	Hairs on head
C	Fifth month	R	Heart
D	Six month	S	Limbs and Digits

(1) A - R, B - S, C - P, D - Q

(2) A - Q, B - R, C - S, D - P

(3) A - S, B - Q, C - R, D - P

✓ (4) A - R, B - S, C - Q, D - P



103. In man the sperm released from the testis take the following route to reach the urethra

- ✓ (1) Vasa efferentia, epididymis, vas deferens and urethra
- (2) Vasa efferentia, Bidder's canal, nephrostome
- (3) Vasa efferentia, collecting tubules and Bidder canals
- (4) Vasa efferentia, Bidder's canal, uriniferous tubules and nephrostome



104. Spermatogonium undergoes ____.

- (1) Reduction division
- ✓ (2) Mitosis
- (3) Meiosis
- (4) None of these



109. Which one is associated with production of androgens?

(1) ACTH

(2) ADH

✓ (3) ICSH

(4) TSH



106. In the given columns, column-I contain various phases of menstrual cycle and column-II contain its features. Select the correct match from the options given below.

Column I		Column II	
A	Menstrual phase →	P	Breakdown of <u>endometrial lining</u> of uterus along with its <u>blood vessels</u> which form <u>liquid</u> that <u>comes out</u> of vagina.
B	Luteal phase →	Q	A temporary endocrine gland is formed and secretes a hormone which maintains endometrium and implantation of fertilized ovum and other events of pregnancy.
C	Follicular phase	R	Secretion of <u>luteinizing hormone</u> at its maximum level and induces breakdown of <u>mature follicle</u> to release the female gamete
D	Ovulatory phase	S	Formation of <u>mature Graafian follicle</u> and regeneration of endometrium of uterus.

- (1) A-S, B-Q, C-P, D-R
 (2) A-P, B-R, C-S, D-Q
 (3) A-P, B-S, C-Q, D-R
 (4) A-P, B-Q, C-S, D-R ✓



107. The wall of uterus consists of

- (1) Inner thin membranous perimetrium: it consists of smooth muscle fibres.
- ✓ (2) Middle myometrium: it consists of smooth muscle fibres.
- (3) Outer endometrium: it consists of striated muscle fibres.
- (4) Middle myometrium: it consists of striated muscle fibres.



108. The seminiferous tubules of the testis open into the vasa efferentia by

- (1) Epididymis
- (2) Seminiferous tubule
- (3) Vas deferens
- ✓ (4) Rete testis



109. Match the columns.

Column I		Column II	
A	Primary oocyte →	P	It is formed when <u>oogonia</u> starts <u>division</u> and <u>temporarily arrested</u> at <u>prophase of meiosis</u>
B	Secondary oocyte	Q	Degenerate <u>during</u> the phase from puberty to birth
C	Primary follicle	R	It is formed <u>after</u> the <u>completion</u> of <u>meiosis-I</u>
D	Secondary follicle →	S	Surrounded by many layers of granulosa cells

(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

✓ (3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



110. Assertion : The sperm head contains an elongated haploid nucleus, the anterior ~~posterior~~ portion of which is covered by a cap-like structure, acrosome. ✗
acw some

Reason : The middle piece is filled with enzymes that help fertilisation of the ovum. ✗

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

✓ (4) Both Assertion and Reason are incorrect



111. Find out which of the following statements are true (T)/ false (F) and choose the correct option.

i. The menstrual phase is followed by the follicular phase. T

ii. The change in the ovary and uterus during menstrual cycle are induced by the changes in the levels of ovarian hormones only. X F

iii. Lack of menstruation may be indicative of pregnancy T

iv. Menstrual flow occurs if the released ovum gets fertilized. F

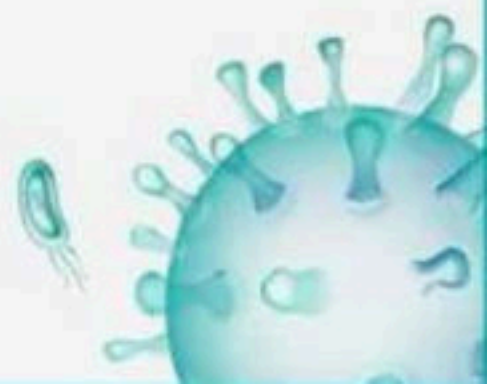
(1) i - T, ii - T, iii - F, iv - F ✓ (2) i - T, ii - F, iii - T, iv - F

(3) i - T, ii - T, iii - T, iv - F (4) i - F, ii - T, iii - T, iv - T



112. The difference between spermatogenesis and oogenesis is?

- (1) Oogenesis is uninterrupted meiosis, but spermatogenesis is arrested at some points
- ☒ (2) Spermatogenesis involves equal meiotic division whereas oogenesis involves unequal meiotic division
- (3) Oogenesis occurs entirely in the ovary whereas spermatogenesis occurs mostly in the testis
- (4) Spermatogenesis is initiated during the embryonic stage itself whereas oogenesis is initiated at menarche



113. The correct sequence of spermatogenetic stages leading to the formation of sperms in mature human testes is

- ☒ (1) Spermatogonia-spermatocytes- spermatids-sperms
- (2) Spermatids-spermatocytes- Spermatogonia- sperms
- (3) Spermatogonia- Spermatids- spermatocytes- sperms
- (4) Spermatocytes- Spermatogonia- Spermatids- sperms



114. Assertion : Graafian follicle ruptures and release ovum (ovulation).

Reason : Rapid secretion of LH leads to its maximum level during the midcycle called LH surge.

- ✓ (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



115. Which of the following hormones attains a peak level in the middle of menstrual cycle?

↓
FSH and LH

- (1) LH and estrogen
- (2) FSH and progesterone
- ✓ (3) FSH and LH
- (4) Estrogen and progesterone



116 Embryonic body form stage is characterized by

- (1) Basic body plan emerging
- ✓ (2) Stage of development resembling other species
- (3) Miniature adult features
- (4) Tissue not fully formed



117. The testis are situated outside the abdominal cavity in scrotum as in helps to



- ✓ (1) Maintain the low temperature
- (2) Store sperm
- (3) Regulates hormone secretion
- (4) Release sperm



118. Which among the following has 23 (haploid) chromosomes?

(1) Spermatogonia

(2) Zygote

☒ (3) Secondary occyte

(4) Oogonia



119. Accessory ducts of reproductive system of human female include:

- (1) Oviduct, uterus and vagina
- (2) Oviduct, ovaries and ovarian ligaments
- (3) Oviduct, ovaries and mammary glands
- (4) Ovaries, uterus and vagina



120. Placenta also acts as a/an ____ tissue.

(1) Exocrine

✓ (2) Endocrine

(3) Paracrine

(4) Merocrine



121. What happens during fertilization when many sperms reach close to ovum

- (1) All sperms except the one nearest to ovum lose their tails
- (2) Cells of corona radiata trap all the sperms except one
- ☒ (3) Secretion of acrosome helps one sperm enter the ovum through zona pellucida
- (4) Only two sperms nearest to ovum penetrate zona pellucida

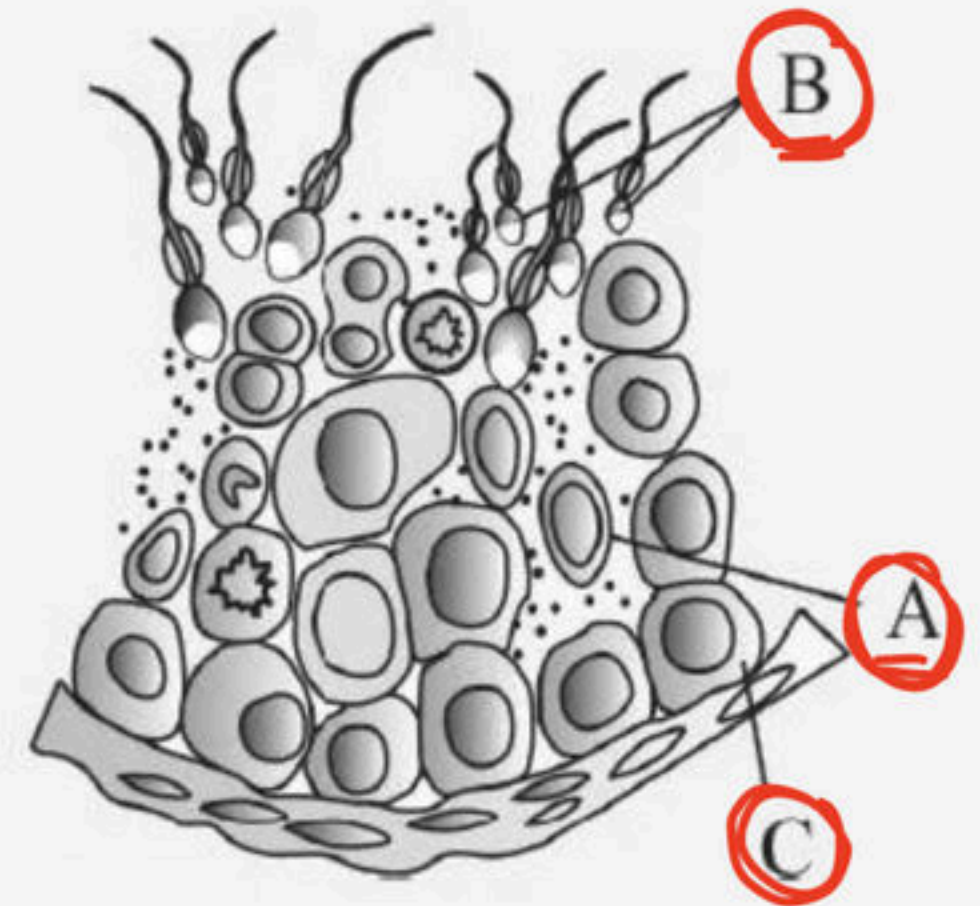


122. Select the correct option.

(1) A - Primary spermatocyte,
B - Sertoli cells,
C - Spermatogonium

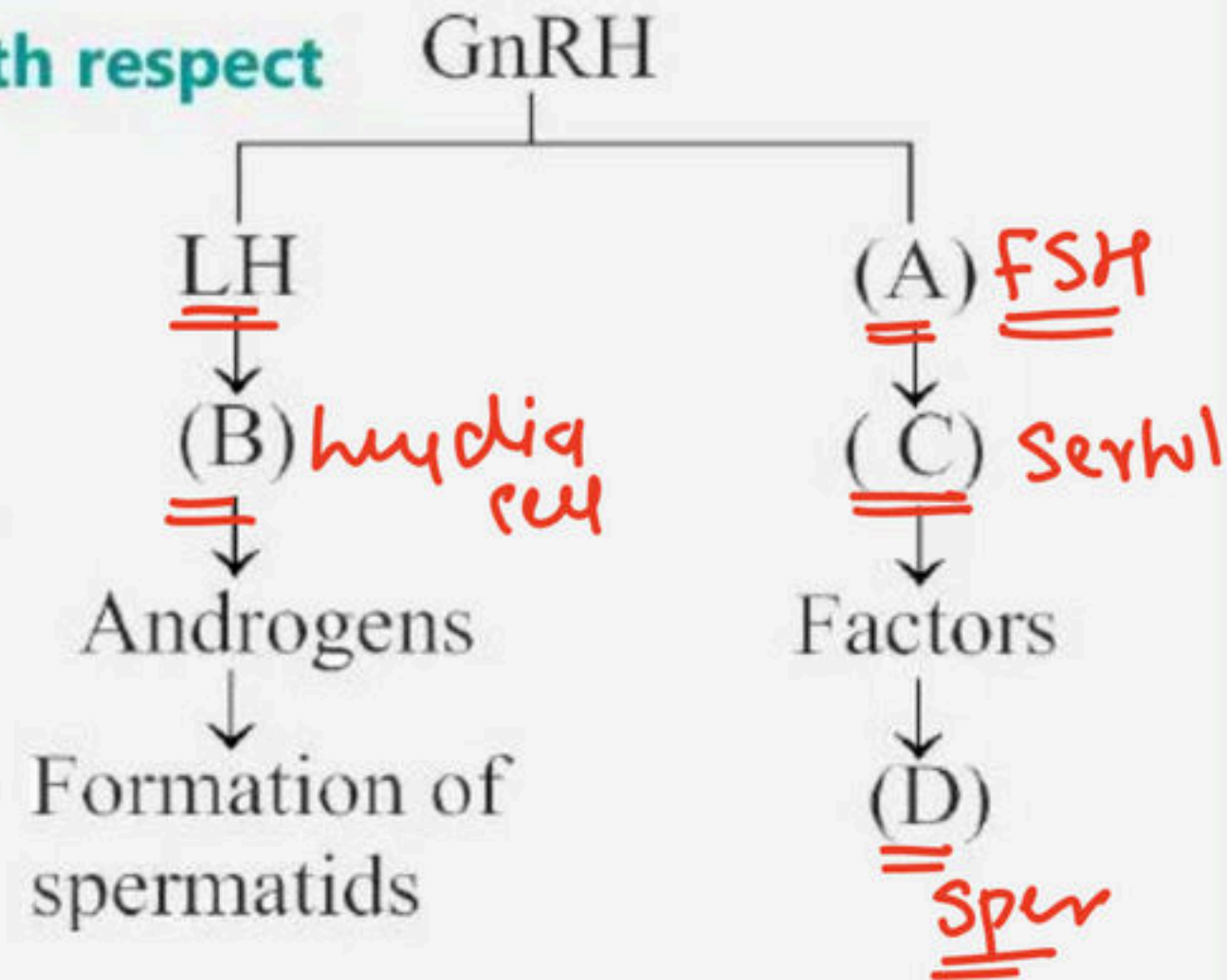
✓ (2) A - Sertoli cells,
B - Spermatozoa,
C - Spermatogonium

(3) A - Sertoli cells,
B - Primary spermatocyte,
C - Spermatozoa
(4) A - Sertoli cells,
B - Spermatozoa,
C - Primary spermatocyte



123. Identify the correct option (A), (B), (C), (D) with respect to spermatogenesis.

- ✓ (1) FSH, Leydig cells, Sertoli cells, spermiogenesis.
(2) ICSH, Interstitial cells, Leydig cells, spermiogenesis.
(3) FSH, Sertoli cells, Leydig cells, spermatogenesis.
(4) ICSH, Leydig cells, Sertoli cells, spermatogenesis.



124. ____ is secreted by the secondary oocyte.

(1) Corona radiata

(2) Corpus luteum

✓✓ (3) Zona pellucida

(4) FSH



125. Which one of the following causes the mammary glands to enlarge at puberty?

- (1) Testosterone
- (2) Progesterone
- ✓ (3) Estrogen
- (4) Oxytocin



126. Seminal plasma in human males is rich in

(1) Ribose and potassium

(2) Glucose and calcium

✓ (3) Fructose and calcium

(4) DNA and testosterone



127. Choose the correct sequence of events occur in human reproduction.

(1) Gametogenesis → Gestation → Insemination →

Fertilization → Implantation → Parturition

(2) Gametogenesis → Insemination → Gestation →

Implantation → Fertilization → Parturition

☒ (3) Gametogenesis → Insemination → Fertilization →

Implantation → Gestation → Parturition

(4) Gestation → Gametogenesis → Insemination →

Implantation → Fertilization → Parturition



128. Which of these is not an important component of initiation of parturition in humans?

- (1) Increase in estrogen and progesterone ratio
- (2) Synthesis of prostaglandins
- (3) Release of oxytocin
- (4) Release of prolactin ✓✓



129. The functional unit of testis of man is

- (1) Uriniferous tubule
- ✓ (2) Seminiferous tubule
- (3) Acini
- (4) Testicular Lobule

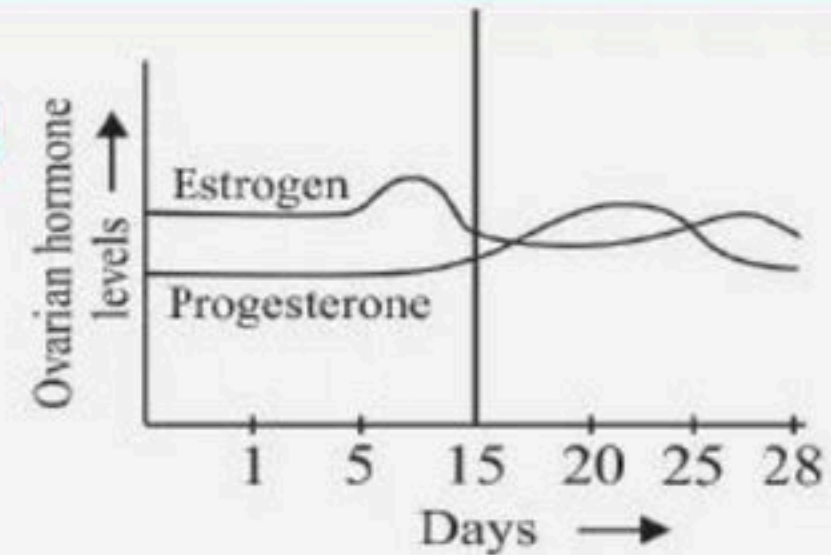


130. Read the graph and correlate the uterine events that take place according to the hormonal levels on

i. 6-15 days →

ii. 16-25 days →

iii. 26-28 days (if the ovum is not fertilized).

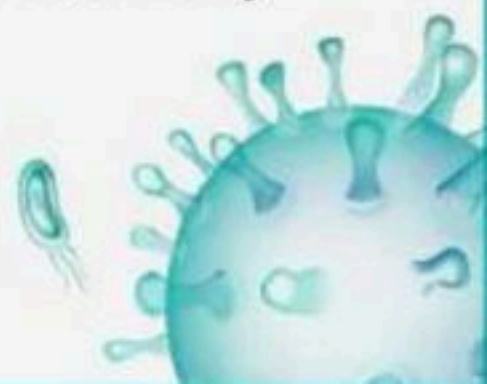


(1) i - Degeneration of endometrium, ii - Endometrium thickens, becomes vascularised, ready to receive and implant ovum, iii- Regeneration of endometrium.

(2) i - Degeneration of endometrium, ii- Myometrium thickens, becomes vascularised, ready to receive and implant embryo, iii- Regeneration of endometrium.

✓ (3) i - Regeneration of endometrium, ii- Endometrium thickens, becomes vascularised, ready to receive and implant embryo, iii- Degeneration of endometrium.

(4) i- Degeneration of endometrium, ii- Endometrium thickens, becomes vascularised, ready to receive and implant embryo, iii- Regeneration of endometrium.



131. Fertilization is defined as the process by which

(1) A diploid spermatozoan unites with a haploid ovum to form a diploid zygote

(2) A diploid spermatozoan unites with a diploid ovum to form a diploid zygote

✓ (3) A haploid spermatozoan unites with a haploid ovum to form a diploid zygote

(4) A diploid spermatozoan unites with a haploid ovum to form a triploid zygote.



132. Acrosomal reaction of the sperm occurs due to

- ✓ (1) Its contact with zona pellucida of the ova
- (2) Reactions within the uterine environment of the male
- (3) Reactions within the epididymal environment of the male
- (4) Androgens produced in the uterus



133. Which of the following statement regarding human development is correct?

(1) Before one month of pregnancy, the embryo's heart is formed.

(2) Chorion develops from inner cell mass ✗

✓ (3) Inner cell mass contains stem cells

(4) By the end of about 20 weeks eye-lids separate



134. Match the columns.

Column I		Column II	
A	Spermatogenesis	P	Release of sperm into female reproductive system
B	Oogenesis	Q	Formation of <u>male gametes</u>
C	Insemination	R	Formation of female gametes
D	Fertilisation	S	Embryonic development
E	Implantation	T	Fusion gametes
F	Gestation	U	Attachment of blastocyst to uterine wall

- (1) A – R, B – P, C – Q, D – T, E – U, F – S
- (2) A – Q, B – R, C – P, D – T, E – U, F – S
- (3) A – S, B – R, C – P, D – T, E – U, F – Q
- (4) A – Q, B – T, C – P, D – R, E – U, F – S



135. Assertion : The secondary oocyte further develops into ovum that undergoes fertilization.

Reason : The secondary oocyte retains bulk of the "nutrient" rich cytoplasm of the primary oocyte.

- ✓ (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



136. At menopause there is rise in urinary excretion of

(1) MSH

✓ (2) FSH

(3) LH

(4) STH



137. Identity the odd one from the following

- ✓ (1) Labia minora ↩
- (2) Fimbriae
- (3) Infundibulum
- (4) Isthmus



138. Release of oocytes/ova from ovary is

(1) Gestation

☒ (2) Ovulation

(3) Parturition

(4) Implantation



139. Spot the odd one out from the following structures with reference to the male reproductive system

- (1) Rete testis
- (2) Epididymis
- (3) Vasa efferentia
- (4) Isthmus



140. Phase of menstrual cycle in human that lasts for 7-8 days is



- ✓ (1) Follicular phase
- (2) Ovulatory phase
- (3) Luteal phase
- (4) Menstruation



141. In human adult females oxytocin

- (1) Stimulates pituitary to secrete vasopressin
- ✓ (2) Causes strong uterine contractions during parturition
- (3) Is secreted by anterior pituitary
- (4) Stimulates growth of mammary glands



142. Statement A : Head of sperm consists of acrosome and mitochondria. nucleus x

Statement B : Acrosome contains a spiral row of mitochondria. x

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- ☒ (4) Both Statements are incorrect.



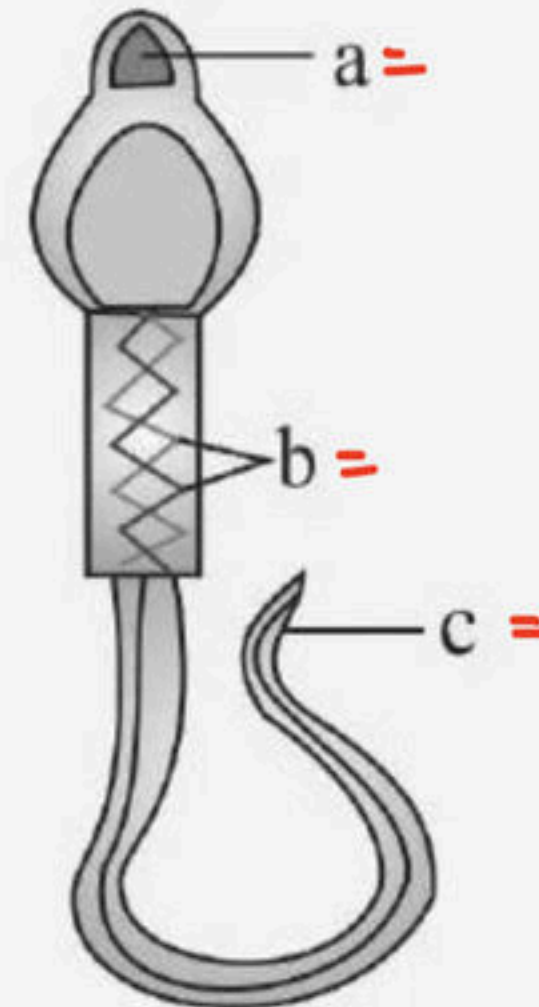
143. In the given diagram of human sperm, identify the functions of the labelled parts, a, b and c.

(1) $a \rightarrow$ Helps in penetration of sperm into ovum
 $b \rightarrow$ Helps in movement of sperm
 $c \rightarrow$ Provides energy for the movement of sperms into the female reproductive tract

✓ (2) $a \rightarrow$ Helps in penetration of sperm into ovum
 $b \rightarrow$ Provides energy for the movement of sperm
 $c \rightarrow$ Helps in movement of sperm

(3) $a \rightarrow$ Helps in movement of sperm
 $b \rightarrow$ Helps in penetration of sperm into ovum
 $c \rightarrow$ Provides energy for the movement of sperms

(4) $a \rightarrow$ Provides energy for the movement of sperm
 $b \rightarrow$ Helps in movement of sperm
 $c \rightarrow$ Helps in penetration of sperm into ovum



144 Capacitation refers to changes in the

- (1) Ovum after fertilization
- (2) Sperm after fertilization
- ✓ (3) Sperm before fertilization
- (4) Ovum before fertilization



145. Which of the following contains a fluid filled cavity called antrum?

- (1) Primary spermatocyte
- (2) Primary follicle
- ✓ (3) Tertiary follicle
- (4) Secondary spermatocyte



146. Which of the following cells present in the human testes forms the sperms?

- (1) Interstitial cells
- ✓ (2) Spermatogonia ✓
- (3) Leydig cells
- (4) Sertoli cells



147. How many of the following structures contain 46 chromosomes?

²ⁿOogonia, ²ⁿSpermatid, Polar body, ²ⁿPrimary oocyte, ²ⁿSpermatozoa, ²ⁿPrimary spermatocyte, Secondary spermatocyte, Spermatogonia, Secondary oocyte.

- (1) Seven
- (2) Eight
- (3) Five
- ☒ (4) Four



148 In humans, secretions of which of the following is used to confirm implantation of embryo?

(1) Blastocyst

(2) Gastrula

✓ (3) Trophoblast → attached to endometrium.

(4) Inner cell mass



149. The difference between primary²ⁿ and secondaryⁿ spermatocyte lies in _____

- (1) Primary gamete/secondary gamete in males
- ✓ (2) Number of chromosomes
- (3) Being hormone producing/non-hormone producing
- (4) Presence/absence of a tail



150. Match the columns.

Column I		Column II	
A	GnRH	P	Stimulates <u>spermatogenesis</u>
B	LH	Q	Acts on <u>anterior pituitary gland</u>
C	Androgens	R	Stimulates <u>secretion of androgens</u>
D	FSH	S	Acts on <u>sertoli cells</u>

(1) A – R, B – P, C – Q, D – S

☒ (2) A – Q, B – R, C – P, D – S

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



151. Hormone having no role in menstruation is

- (1) LH
- (2) FSH
- ✓(3) GH
- (4) None of these



152. Choose the wrong statement :-

- I. Number of autosomes in 1^{2n} and 2^n spermatocytes are equal. ✗
- II. Number of allosomes in spermatogonium and 1^n spermatocytes is equal. ✓
- III. Number of autosomes in spermatid & sperm is not equal. ✗
- IV. Number of autosomes in 2^n spermatocyte and sperm is equal. ✓
- V. Number of pairs of autosomes in 2^n spermatogonia is equal to that in 1^{2n} spermatocyte. ✗

(1) I, II

(2) II, III, IV

(3) I, II, IV

✓ (4) I, III, V



153. In human females at the time of birth there are two million ova, how many of them normally reach maturity incourse of normal reproductive life (approximately)

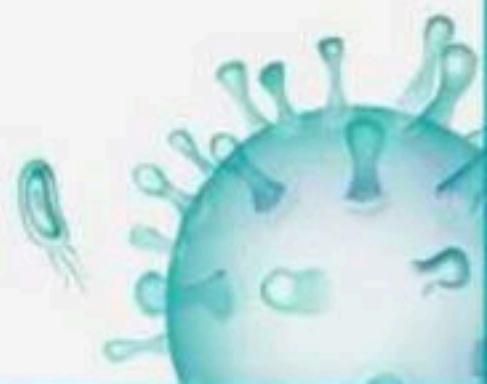
✓ (1) 500

300-500

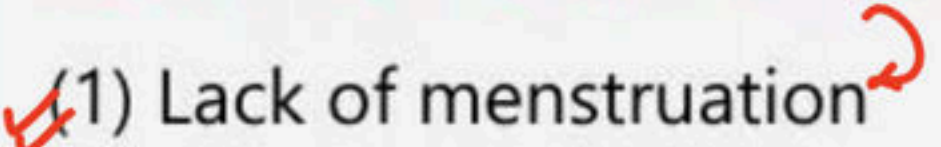
(2) 1,000

(3) 2,000

(4) 5,000



154. Which of the following is associated with pregnancy?

- ✓ (1) Lack of menstruation 
- (2) Occurrence of menstrual flow
- (3) Low levels of estrogen
- (4) Low levels of progesterone



155. Which one of the following is most likely root cause why menstruation is not taking place in regularly cycling human female?

- (1) Maintenance of the hypertrophical endometrial lining
- (2) Maintenance of high concentration of sex hormones in the blood stream
- (3) Retention of well-developed corpus luteum
- ✓ (4) Fertilization of the ovum



156. Prostate gland is present. :-

- (1) On ureter
- (2) On the urinary bladder
- (3) Below testis
- ☒ (4) Below the urinary bladder ☒



157. The funnel shaped part of fallopian tube which traps ovum is known as

- (1) Ampulla
- (2) Isthmus
- ✓ (3) Infundibulum
- (4) fimbriae



158 Choose the incorrect pair.

- (1) Finger-like projections - Infundibulum
- ✓ (2) Narrow part of oviduct - Ampulla = widder
- (3) Part of oviduct joining the uterus - Isthmus
- (4) Epithelium of fallopian tube - ciliated



159. The hormone that shows two peaks in one menstrual cycles also causes

- (1) Maintenance of endometrium
- (2) Ovulation
- (3) Release of FSH from pituitary
- ✓ (4) Female secondary sexual characters



160. The cycle of events starting from one menstruation till the next one is called

- (1) Pregnancy
- (2) Parturition
- (3) Implantation
- ✓ (4) Menstrual cycle ✓



161. Assertion : Blastocyst becomes embedded in the endometrium of the uterus. ✓

Reason : After attachment of inner cell mass to the endometrium, the uterine cells divide rapidly and covers the blastocyst. ✗

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

✓(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect



162. Fertilization in humans is practically feasible only if. :-

- (1) The sperms are transported into vagina just after the release of ovum in fallopian tube
- (2) ✓✓ The ovum and sperms are transported simultaneously to ampullary ~~isthmic~~ junction of the fallopian tube
- (3) The ovum and sperms are transported simultaneously to ampullary - isthmic junction of the cervix
- (4) The sperms are transported into cervix within of release of ovum in uterus



163. Menstrual phase is followed by

(1) Luteal phase

✓ (2) Follicular phase

(3) Fertilization

(4) Implantation



164. Select the correct sequence for transport of sperm cells in male reproductive system.

- (1) Testis → Epididymis → Vasa efferentia → Rete testis → Inguinal canal → Urethra
- ☒ (2) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra → Urethral meatus
- (3) Seminiferous tubules → Vasa efferentia → Epididymis → Inguinal canal → Urethra
- (4) Testis → Epididymis → Vasa efferentia → Vas deferens → Ejaculatory duct → Inguinal canal → Urethra → Urethral meatus



165. Which part of the sperm provides energy to fertilize the egg?

- (1) Sperm head
- (2) Acrosome
- (3) Sperm tail
- (4) ☒ Middle piece



166. Secondary follicles can be distinguished from primary follicles with help of ____.

- (1) Granulosa cells covering.
- ✓ (2) Presence of theca layer
- (3) Presence of antrum.
- (4) All of these



167. How many of the following hormones noticed in the blood of a pregnant woman?

Progest^①erone, est^②rogen, hC^③G, hP^④L, Relaxin^⑤ cortisol^⑥

(1) 2

(2) 3

(3) 4

✓ (4) 6



168. In amniocentesis, the fluid for test is taken from

- (1) Foetal blood
- (2) Mother's blood
- ✓ (3) Amniotic fluid
- (4) Mother's body fluid



169. Which of the following hormone levels will cause release of ovum (ovulation) from the Graffian follicle?

- (1) High concentration of progesterone
- (2) Low concentration of LH
- (3) Low concentration of FSH
- (4) ☒ High concentration of Estrogen ↑



170. Match the column-I with column-II.

Column I		Column II	
A	Semiferous tubules	P	Formation of testosterone
B	Sertoli cells	Q	<u>Spermatogenesis</u>
C	Leydig cells	R	Storage of sperms
D	Epididymis	S	Transport of sperms
E	Vasa deferens	T	Nourishment of sperms

- (1) A – R, B – P, C – Q, D – T, E – S
(2) A – Q, B – R, C – P, D – T, E – S
(3) A – S, B – R, C – P, D – T, E – Q
☒ (4) A – Q, B – T, C – P, D – R, E – S



171. Spermiation is the process of the release of sperms
from



✓ (1) Seminiferous tubules ✓

(2) Vas deferens

(3) Epididymis

(4) Prostate gland



172. How many eggs and sperms are required to create identical twins in humans?

→ 2 yoke → 0
✓ (1) Egg – 1, Sperms – 1

(2) Egg – 2, Sperms – 2

(3) Egg – 2, Sperms – 1

(4) Egg – 1, Sperms – 2



173. Study the given process.

Secondary Oocyte $\xrightarrow[\text{division}]{2^{\text{nd}} \text{ meiotic}}$ ovum + 2nd polar body Identify

correctly, when does the given division occurs.

- (1) After spermiogenesis.
- ✓ (2) After fusion of sperm and ovum.
- (3) After ovulation.
- (4) After sperm enters the ampulla of oviduct.



174. In an ideal menstrual cycle, the menstrual phase last
for



- ✓ (1) 3-5 days
- (2) 15-16 days
- (3) 10 days
- (4) 12-21 days



175. A long, formed by union of seminal duct and the terminal end of vas deferens opens into urethra.

- (1) Vas deferens
- (2) Vas efferens
- ☒ (3) Ejaculatory duct
- (4) Rete testis



176. Which is wrong about oogenesis?

- ✓ (1) Equal division ✗
- (2) Unequal division
- (3) Formation of polar bodies
- (4) Growth phase



177. What is the size of testis?

- (1) 2 to 3 cm in length and 4 to 5 cm in width
- ☒ (2) 4 to 5 cm in length and 2 to 3 cm in width
- (3) 6 to 7 cm in length and 4 to 5 cm in width
- (4) 4 to 5 cm in length and 6 to 7 cm in width



178 Finger like projection, called chorionic villi, appear on the ____ after the implantation.

(1) Ampulla

✓(2) Trophoblast

(3) Infundibulum

(4) Inner cell mass



179. Spermiogenesis is

- (1) Transformation of sperm into spermatids.
- (2) Transformation of spermatogonia into primary spermatocyte.
- (3) Transformation of secondary spermatocyte into spermatids.
- ✓ (4) Transformation of spermatid into spermatozoa.



180. Which is the correct path for the secretion and transport of milk in mammary gland?

- (1) Mammary duct → Cavity of alveoli → Lactiferous duct
→ Mammary ampulla → Mammary tubule → Cavity of alveoli
- (2) Mammary duct → Mammary tubule → Alveoli →
Lactiferous duct → Mammary ampulla → Cavity of alveoli
- ☒ (3) Alveoli → Cavity of alveoli → Mammary tubule →
Mammary duct → Mammary ampulla → Lactiferous duct
- (4) Alveoli → Mammary tubule → Mammary ampulla →
Cavity of alveoli → Mammary duct → Lactiferous duct



181. Which gland in female is a counterpart of Cowper's gland in male?

- (1) Perineal gland
- (2) Clitoris
- ✓ (3) Bartholin's gland
- (4) None of these



182. Study the following statements and answer the question. During ' ' ^{luteal} phase of the menstrual cycle, if pregnancy doesn't happen, the 'Q' ^{C.L} withers and dies. Usually around day 22 in a 28 day cycle. The drop in ' ' ^{prog} levels causes the lining of the uterus to fall away. This is known as 'S'. ^{men} Identify P, Q, R and S.

- (1) P-Menstrual; Q-Graafian follicle; R-Estrogen; S-Menarche
- (2) P-ovulatory; Q-endometrium; R-Follicle stimulating hormone; S-Menopause
- ✓ (3) P-Luteal; Q-Corpus luteum; R-progesterone; S-Menstruation
- (4) P-Follicular; Q-secondary oocyte; R Luteinizing hormone; S-Menstruation



183. Name the primary sex organ and the endocrine gland in the males

(1) Testis, epididymis

✓ (2) Testis, Testis

(3) Testis, Vasa deferens

(4) Testis, Rete testis



184. Human female has 44+XX chromosomes. The secondary oocyte shall have

(1) 44+XX

✓(2) 22+X

(3) 22

(4) 44



185. Which is not the accessory duct in male reproductive system of human?

(1) Vasa efferentia

✓ (2) Seminiferous tubule

(3) Epididymis

(4) Vas deferens



186. Blastomeres are daughter cells formed in the process of

- ✓ (1) Cleavage, when zygote undergoes mitotic division.
- (2) Fertilization, when sperm enters in the cytoplasm of ovum.
- (3) Implantation, when blastocysts attached to the uterine endometrium.
- (4) Gametogenesis, when male and female gametes are produced by testis and ovary respectively.



187. How many ova and sperms would be produced from 100 secondary oocytes and 100 secondary spermatocytes during gametogenesis in human?

100 ova

200

(1) 100 ova, 100 sperms

☒ (2) 100 ova, 200 sperms

(3) 50 ova, 100 sperms

(4) 200 ova, 200 sperms



188. After insemination the correct pathway of sperm movement is

- (1) Vagina → Cervix → Uterus → Ampulla → Isthmus
- (2) Vagina → Uterus → Cervix → Isthmus → Ampulla
- (3) Vagina → Cervix → Ampulla → Uterus → Isthmus
- (4) ✓ Vagina → Cervix → Uterus → Isthmus → Ampulla



189. Which group represents the external genitalia

- ✓ (1) Labium minora, labium majora and clitoris
- (2) Labium minora, labium majora and vagina
- (3) Labium minora, labium majora and cervix
- (4) Labium minora, labium majora and oviduct



190. For a normal male fertility, which of the following statements is correct?

- (1) 60 percent sperm must have normal motility and 40 percent must have normal shape.
- ✓ (2) 60 percent sperm must have normal shape 40 percent must have vigorous motility.
- (3) 60 percent sperm must have normal shape and 40 percent must have acrosome.
- (4) None of these



191. Which of the following statements about ovary is not true?

- (1) Ovaries are responsible for the production of several steroid hormones.
- ☒ (2) Each ovary is about ²⁻⁴~~4~~ to ~~6~~ cm in length.
- (3) Ovary is connected to the pelvic wall.
- (4) Ovary produces female gamete (Ovum).



192. Menstrual flow occurs due to lack of

- (1) Oxytocin
- (2) Vasopressin
- ✓ (3) Progesterone
- (4) FSH



193. The acrosome found in a sperm is made up of

(1) Lysosome

(2) Centriole

✓ (3) Golgi bodies ✓

(4) Mitochondria



194. In human females, the endometrium of uterus consists of

- (1) Smooth muscle
- ✓✓ (2) Glandular layer
- (3) Adipose layer
- (4) Cartilaginous layer

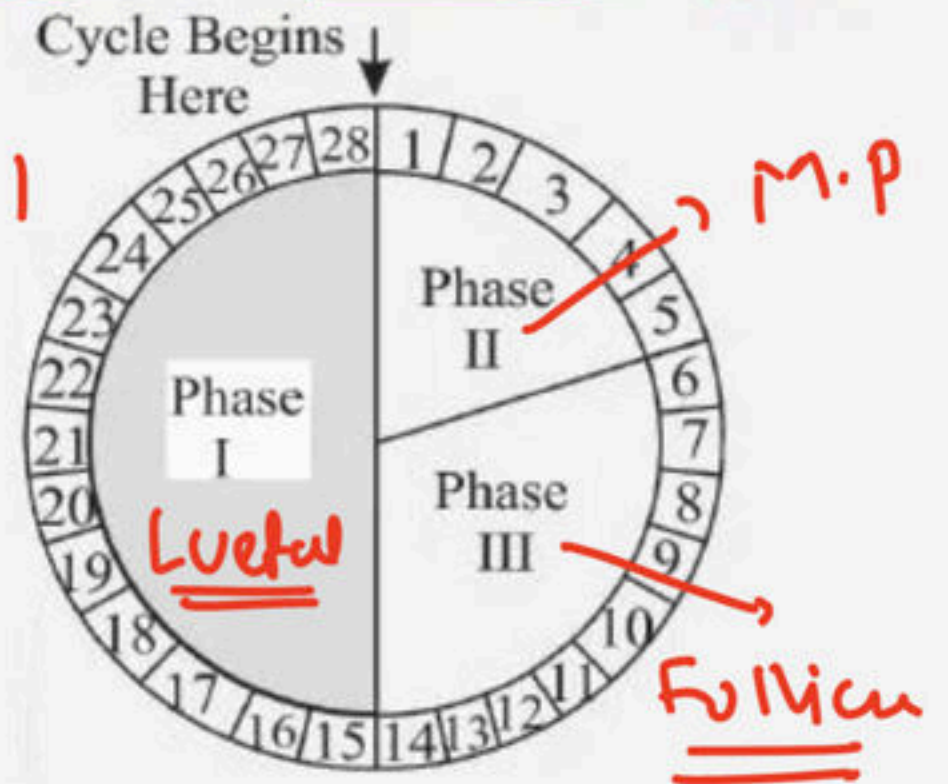


195. Refer the given schematic representation of menstrual cycle showing three phases, i, ii, iii. Select the option that correctly matches these phases with the following events.

i. FSH secreted by the anterior pituitary stimulates the ovarian follicle to secrete oestrogen which stimulates the proliferation of the endometrium. ✓

ii. LH secreted by the anterior pituitary stimulates the development of corpus luteum which secretes progesterone. → Phase I

iii. Reduced production of LH from anterior pituitary causes degeneration of corpus luteum, thereby progesterone production is reduced. → M.P?



(1) Phase I – iii, Phase II – i, Phase III – ii

(2) Phase I – i, Phase II – ii, Phase III – iii

(3) Phase I – iii, Phase II – ii, Phase III – i

✓ (4) Phase I – ii, Phase II – iii, Phase III – i

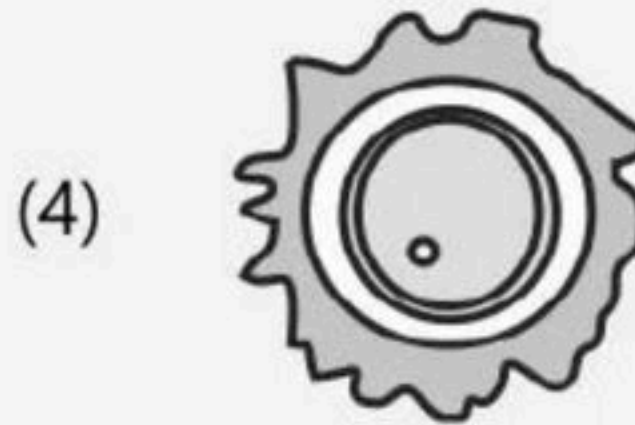
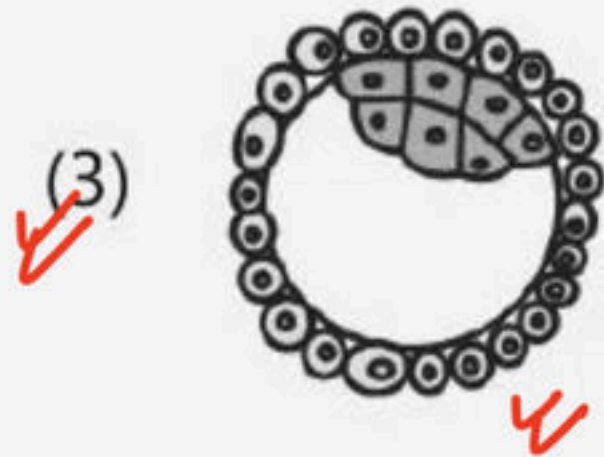
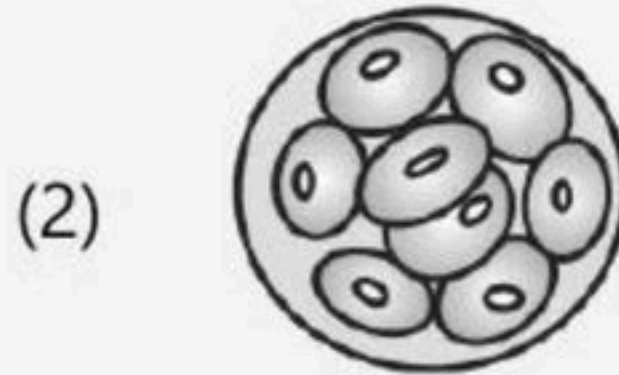
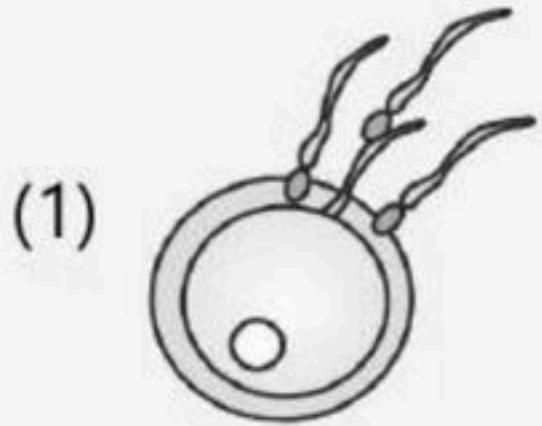


196. What happens during fertilization when many sperms reach close to ovum

- (1) Cells of corona radiata trap all the sperm except one
- (2) Only two sperms nearest to ovum penetrate zona pellucida
- ☒ (3) Secretion of acrosome helps one sperm enter cytoplasm of ovum through zona pellucida
- (4) All sperms except the one nearest to ovum lose their tails



197. Which of the following human developmental stage becomes embedded in the uterine endometrium by a process called implantation and leads to pregnancy?



198 Each seminiferous tubule has two types of cells:

- (1) Female germ cells (spermatogonia) and Sertoli cells.
- ✓ (2) Male germ cells (spermatogonia) and Sertoli cells.
- (3) Male germ cells (spermatogonia) and Leydig cells
- (4) Female germ cells (spermatogonia) and Leydig cells.



199. Select the correct sequence:

- (1) GnRH → LH → Leydig cells → Androgens → Spermatogenesis
- (2) GnRH → FSH → Sertoli cells → Some factors → Spermatogenesis
- (3) Both are incorrect
- (4) Both are correct
- Handwritten notes:*
- Arrows from "formation" point to "Androgens" and "Spermatogenesis".
 - A bracket groups "Spermatogenesis" in (1) and (2) with the word "Spermatid".
 - A bracket groups "Spermatogenesis" in (2) with the word "Spermiogen".
 - An "L," is written below "Some factors".



200. Assertion : The presence or absence of hymen is a reliable indicator of virginity. X

Reason : The hymen always torn during the first coitus (intercourse). X

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect



THANK YOU



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